

Comprehensive Numerical and Perturbative Analysis of Projectile Motion: Within-System Comparison of Linearized Perturbation and Non-linear Quadratic Drag

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Abstract—This paper presents a comparative study of projectile motion by examining two distinct physical systems under two different mathematical paradigms: a rigid uniform slender rod and a heavy spherical ball (shot put). Rather than performing a cross-body comparison, this study implements a within-system analysis for each object, contrasting a high-fidelity fourth-order Runge-Kutta (RK4) numerical integration of non-linear quadratic drag against a first-order analytical perturbation approximation that linearizes the drag force at the launch velocity. For the rigid rod, the RK4 formulation additionally incorporates coupled aerodynamic torque. Our results demonstrate that the linearized perturbation model remains exceptionally accurate for highly dense projectiles with low smallness parameters, whereas it begins to deviate for lighter, high-area geometries under high launch velocities.

Index Terms—Projectile dynamics, Perturbation method, Runge-Kutta 4, Aerodynamic torque, Rigid body rotation, Linearization.

I. INTRODUCTION

The study of projectile motion serves as a foundational pillar in classical mechanics, bridging theoretical kinematics with empirical ballistics [6]. In elementary physics, atmospheric effects are often neglected, yielding symmetric parabolic trajectories governed entirely by uniform gravitational fields. However, real-world applications—ranging from ballistics and aerospace engineering to the dynamics of wind-blown debris—demand the integration of non-linear environmental forces, such as velocity-dependent aerodynamic drag and rotationally-induced torques.

When atmospheric resistance is incorporated, the governing ordinary differential equations (ODEs) frequently lose their closed-form analytical solvability. While a linear drag model $F_d \propto v$ permits an exact logarithmic solution, a quadratic drag model $F_d \propto v^2$ results in coupled non-linear equations that necessitate approximate analytical techniques or iterative numerical methods. One powerful analytical alternative is the perturbation method, which expands the state variables in a power series of a small dimensionless parameter representing weak environmental friction [1].

In this paper, we develop a rigorous framework to explore these phenomena. Unlike typical studies that compare different

objects, we construct a customized Python simulator that evaluates each system (either the rod or the ball) against itself using two competing models: (1) a full non-linear quadratic drag model solved numerically via RK4, and (2) a linearized first-order perturbation model. For the rod, the numerical model also includes a coupled aerodynamic torque. This within-system comparison strictly isolates the mathematical differences between quadratic velocity coupling and linearized decoupling.

II. CRITICAL REVIEW OF RELEVANT LITERATURE

The transition from idealized vacuum ballistics to realistic atmospheric flight requires a nuanced understanding of aerodynamic drag models and rigid body dynamics. The domains of validity for linear versus quadratic drag depend strictly on the Reynolds number (Re) of the flow field, defined as $Re = \rho v \ell / \mu$, where ℓ is a characteristic length and μ the dynamic viscosity of the fluid. Stokes drag, where the resistive force is linearly proportional to the velocity vector, applies only at highly constrained physical extremes ($Re < 1$). For macroscopic objects moving at standard ballistic speeds, however, the drag force scales quadratically with the velocity magnitude due to the dominance of inertial forces [2].

The unsteady aerodynamics of freely falling plates and tumbling rods has been investigated extensively at intermediate Reynolds numbers ($Re \approx 10^3$). Such studies show that the fluid circulation around tumbling bodies is dominated by a rotational term proportional to the angular velocity, which challenges older quasi-steady aerodynamic models. Experimental and computational fluid dynamics work has likewise documented the highly transient interactions between cylinder wakes and downstream aerodynamic structures, underscoring how sensitive slender-body flight is to its instantaneous orientation.

Regarding the generation of aerodynamic torque on slender bodies, Munk's potential-flow formulation established that the pitching moment coefficient for a slender body is directly proportional to $\sin(2\phi)$, where ϕ is the angle of attack [4]. To account for flow separation at higher angles of attack, Allen and Perkins later modified this theory by introducing a viscous

crossflow term [5]. This semi-empirical approach successfully predicts normal forces and pitching moments across wide ranges of Mach numbers and extreme angles of attack.

III. THEORETICAL FRAMEWORK AND DERIVATIONS

A. Quadratic Drag Linearization and Perturbation Formulation

To establish a direct, mathematically rigorous comparison between the numerical quadratic drag model and the analytical perturbation model, we must linearize the quadratic drag force at the launch speed v_0 . The general macroscopic quadratic drag force is written as:

$$\vec{F}_q = -\frac{1}{2}\rho C_d A v \vec{v} \quad (1)$$

where $v = |\vec{v}| = \sqrt{\dot{x}^2 + \dot{y}^2}$ is the instantaneous speed, ρ is the ambient air density, C_d the dimensionless drag coefficient, A the reference cross-sectional area, and m the mass of the projectile. If we approximate the speed $v \approx v_0$ near the initial launch phase, the drag force can be linearized as:

$$\vec{F}_{lin} = -\left(\frac{1}{2}\rho C_d A v_0\right) \vec{v} \quad (2)$$

Expressing this as a linear drag per unit mass, we write $\vec{F}_{lin} = -m\gamma\vec{v}$, where the effective linear drag coefficient γ (units of s^{-1}) is given by:

$$\gamma = \epsilon v_0 \quad \text{where} \quad \epsilon = \frac{\rho C_d A}{2m} \quad (3)$$

Here, ϵ represents the object-specific perturbation parameter (units of m^{-1}). Under this linearized formulation, the equations of motion for the horizontal coordinate x and vertical coordinate y decouple, with g the gravitational acceleration:

$$\ddot{x} = -\epsilon v_0 \dot{x} \quad (4)$$

$$\ddot{y} = -g - \epsilon v_0 \dot{y} \quad (5)$$

Assuming $\epsilon v_0 \ll 1$, we expand $x(t)$ and $y(t)$ in powers of ϵ :

$$x(t) = x_0(t) + \epsilon x_1(t) + \mathcal{O}(\epsilon^2) \quad (6)$$

$$y(t) = y_0(t) + \epsilon y_1(t) + \mathcal{O}(\epsilon^2) \quad (7)$$

The zeroth-order system (ϵ^0) represents the ideal vacuum trajectory, with launch-velocity components $v_{0x} = v_0 \cos \theta_0$ and $v_{0y} = v_0 \sin \theta_0$ for a launch angle θ_0 :

$$x_0(t) = v_{0x}t, \quad y_0(t) = v_{0y}t - \frac{1}{2}gt^2 \quad (8)$$

Collecting terms at first order (ϵ^1) yields the linearized atmospheric correction, driven by the zeroth-order velocities:

$$\dot{x}_1 = -v_0 \dot{x}_0, \quad \dot{y}_1 = -v_0 \dot{y}_0 \quad (9)$$

Substituting $\dot{x}_0 = v_{0x}$ and $\dot{y}_0 = v_{0y} - gt$, and integrating twice under zero initial perturbation conditions ($x_1(0) = \dot{x}_1(0) = y_1(0) = \dot{y}_1(0) = 0$) yields:

$$x_1(t) = -\frac{1}{2}v_0 v_{0x} t^2 \quad (10)$$

$$y_1(t) = -v_0 \left(\frac{1}{2}v_{0y}t^2 - \frac{1}{6}gt^3 \right) \quad (11)$$

Combining these terms, the complete first-order perturbation trajectory equations are:

$$x(t) \approx v_{0x}t - \frac{1}{2}\epsilon v_0 v_{0x} t^2 \quad (12)$$

$$y(t) \approx v_{0y}t - \frac{1}{2}gt^2 - \epsilon v_0 \left(\frac{1}{2}v_{0y}t^2 - \frac{1}{6}gt^3 \right) \quad (13)$$

B. System I: Spherical Ball (Shot Put) Equations of Motion

For the heavy spherical ball, we utilize parameters matching a standard athletic shot put. The numerical model assumes a point-mass sphere experiencing pure quadratic drag without lift or torque, so the Cartesian acceleration components are:

$$a_x = -\frac{\rho C_d A v v_x}{2m}, \quad a_y = -g - \frac{\rho C_d A v v_y}{2m} \quad (14)$$

where $v_x = \dot{x}$ and $v_y = \dot{y}$ are the velocity components, $v = \sqrt{v_x^2 + v_y^2}$ is the speed, and $A = \pi r^2$ is the frontal cross-sectional area of a sphere of radius r .

C. System II: Rigid Slender Rod Equations of Motion

The rigid uniform slender rod of mass m , length L , and cross-sectional reference area A translates and rotates in a single plane. Let θ denote the rod's pitch (orientation) angle and $\omega = \dot{\theta}$ its angular velocity. Because the surrounding air is still, the velocity of the centre of mass relative to the air equals its ground velocity, $\vec{v}_{rel} = \vec{v}$, with components $v_{rel,x} = v_x$ and $v_{rel,y} = v_y$, and magnitude $v_{rel} = \sqrt{v_{rel,x}^2 + v_{rel,y}^2}$. The instantaneous flight-path angle is $\phi_v = \text{atan2}(v_{rel,y}, v_{rel,x})$, so the aerodynamic angle of attack is $\phi = \theta - \phi_v$.

The translational accelerations of the centre of mass obey the same quadratic drag law as the sphere:

$$a_{x,cm} = -\frac{\rho C_d A v_{rel} v_{rel,x}}{2m} \quad (15)$$

$$a_{y,cm} = -g - \frac{\rho C_d A v_{rel} v_{rel,y}}{2m} \quad (16)$$

Following Munk's slender-body theory [4], the aerodynamic pitching moment about the centre of mass is modelled as proportional to $\sin(2\phi)$, the dynamic pressure $\frac{1}{2}\rho v_{rel}^2$, the reference area A , and the rod length L acting as the effective moment arm:

$$\tau = -\frac{1}{2}\rho C_d A L v_{rel}^2 \sin(2\phi) \quad (17)$$

With the slender-rod moment of inertia $I = \frac{1}{12}mL^2$, the angular acceleration $\alpha = \dot{\omega} = \tau/I$ reduces to:

$$\alpha = \frac{\tau}{I} = -\frac{6\rho C_d A v_{rel}^2 \sin(2\phi)}{mL} \quad (18)$$

IV. NUMERICAL METHODOLOGY AND SYSTEM IMPLEMENTATION

A. Fourth-Order Runge-Kutta (RK4) Integration

The state vectors are integrated discretely using the classical fourth-order Runge-Kutta (RK4) scheme [3] with a fixed time step of $\Delta t = 0.01$ s. For the rod, the state vector is $\mathbf{S}(t) = [x, y, v_x, v_y, \theta, \omega]^T$, while the ball uses a reduced four-dimensional vector $\mathbf{S}(t) = [x, y, v_x, v_y]^T$. Each step advances the state as

$$\mathbf{S}_{n+1} = \mathbf{S}_n + \frac{\Delta t}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4), \quad (19)$$

where the four slope evaluations $\mathbf{k}_i$ are produced by a single vectorized derivative routine $\dot{\mathbf{S}} = \mathbf{f}(t, \mathbf{S})$. For the rod, every evaluation of $\mathbf{f}$ recomputes the air-relative velocity $\vec{v}_{rel} = \vec{v} - \vec{v}_{wind}$, the flight-path angle $\phi_v = \text{atan2}(v_{rel,y}, v_{rel,x})$, the angle of attack $\phi = \theta - \phi_v$, and the resulting drag force and pitching moment, so that the translational and rotational degrees of freedom remain fully coupled at every sub-step.

B. Smallness Parameter and Validity Warning System

To evaluate the mathematical validity of the linearized perturbation expansion, we define the unperturbed flight time as $T_{vac} = \frac{2v_0 \sin \theta_0}{g}$. The weak-drag assumption requires the smallness parameter to satisfy:

$$\epsilon v_0 T_{vac} \ll 1 \quad (20)$$

The simulator evaluates this parameter at startup, printing a warning if $\epsilon v_0 T_{vac} \geq 0.3$. This ensures that the user is immediately aware of potential inaccuracies in the first-order Taylor expansion before rendering the trajectories.

C. Simulator Implementation, Termination, and Post-Processing

The simulator is implemented in Python using the NumPy array library, with a common derivative routine wrapped by the RK4 stepper for both bodies. A uniform wind field $\vec{v}_{wind}$ is supported through the air-relative velocity, but it is set to zero for all runs reported here so that the only difference between the two curves is the integration method rather than the environment. To avoid a singularity in the drag direction, the aerodynamic terms are applied only when $|\vec{v}_{rel}| > 0$.

Numerical integration proceeds until the centre-of-mass height first becomes negative (ground impact) or a safety cap of 5000 steps is reached, whichever occurs first. The range and flight time are then read from the last above-ground sample, and the peak height is taken as the maximum of the vertical coordinate over the recorded trajectory. By contrast, the closed-form perturbation solution of Section III is evaluated analytically on a uniform time grid extending to $1.3T_{vac}$; samples for which $y < 0$ are masked so that only the airborne portion is retained.

For the rod, two rotational diagnostics are extracted from the discrete orientation history. The net orientation change is computed as $\Delta\theta = |\theta_{end} - \theta_0|$, whereas the total absolute angular path is accumulated as $\sum_n |\theta_{n+1} - \theta_n|$, from which

an equivalent revolution count is obtained by dividing by 2π . Because this sum is direction-sensitive, it captures every oscillatory reversal of the pitch angle and not merely the net displacement, which is why its revolution-equivalent value can exceed the modest net change $\Delta\theta$.

V. RESULTS AND CRITICAL COMPARATIVE ANALYSIS

The execution of the within-system simulation reveals profound kinematic deviations, starkly demonstrating the physical consequences of linearizing a quadratic drag profile. The core simulation runs are evaluated under identical environmental conditions with standard sea-level atmospheric density ($\rho = 1.225 \text{ kg/m}^3$) and still air, under a gravitational acceleration of $g = 9.81 \text{ m/s}^2$, initialized at a launch speed of $v_0 = 35.0 \text{ m/s}$ and a launch angle of $\theta_0 = 45^\circ$.

A. Rod Mode Analysis and Smallness Violation

In Rod Mode, the physical parameters of the slender rod ($m = 2.0 \text{ kg}$, $L = 1.0 \text{ m}$, $C_d = 0.4$, $A = 0.05 \text{ m}^2$) yield an object-specific perturbation parameter of $\epsilon = 0.006125 \text{ m}^{-1}$. Under the unperturbed flight time of $T_{vac} = 5.0456 \text{ s}$, the resulting smallness parameter is computed as $\epsilon v_0 T_{vac} = 1.0817$. Because this value significantly exceeds the 0.3 threshold, the simulator triggers a validity warning, indicating that the weak-drag assumption required for a first-order Taylor series expansion is severely violated.

The ballistic trajectory metrics in Table I reveal a massive range degradation for the numerical solution and an even more severe under-prediction by the analytical approximation. The high-fidelity RK4 numerical simulation under quadratic drag ($F_d \propto v^2$) yields a maximum range x_{max} of 80.9163 m, a total flight time t_f of 4.4400 s, and a peak altitude y_{max} of 24.2298 m.

In contrast, the first-order perturbation model under-predicts these values significantly, yielding a range of 56.6398 m, a flight time of 4.0257 s, and a peak height of 22.1446 m. This results in a substantial range error of 24.2765 m, representing a 30.00% discrepancy relative to the RK4 baseline. The physical explanation for this discrepancy lies in the nature of the linearization. To match the models at launch, the perturbation solution applies a constant, high effective linear drag coefficient ($\gamma = \epsilon v_0 = 0.2144 \text{ s}^{-1}$) over the entire flight. However, under the true quadratic drag integrated by RK4, the decelerating force is proportional to the square of the velocity (v^2). As the rod climbs and slows down toward the apex, the quadratic drag decays rapidly, allowing the rod to preserve its velocity and glide significantly further than the linearized approximation.

Furthermore, the rotational analysis highlights the coupled motion of the rod, illustrated in Fig. 1. Fired with an initial tilt of 30° , the rod starts with an angle of attack of $\phi = -15^\circ$ relative to the velocity vector. Because the modelled pitching moment $\tau \propto -\sin(2\phi)$ acts to reduce the angle of attack, it is restoring rather than destabilising: as shown in Fig. 1(b), the rod does not tumble end over end but instead executes

TABLE I
SIMULATION METRICS AND WITHIN-SYSTEM COMPARATIVE ANALYSIS

Parameter & Metric	Vacuum Limit	RK4	Perturbation	Discrepancy (RK4 vs. Pert.)
Rod Mode ($\epsilon = 0.006125 \text{ m}^{-1}$)				
$\text{Smallness} = \epsilon v_0 T_{vac} = 1.0817 \geq 0.3$			Invalid	
Max Range x_{max} (m)	124.8726	80.9163	56.6398	24.2765 (30.00%)
Flight Time t_f (s)	5.0456	4.4400	4.0257	0.4143 (9.33%)
Peak Height y_{max} (m)	31.2181	24.2298	22.1446	2.0853 (8.61%)
Ball Mode ($\epsilon = 0.000448 \text{ m}^{-1}$)				
$\text{Smallness} = \epsilon v_0 T_{vac} = 0.0792 < 0.3$			Valid	
Max Range x_{max} (m)	124.8726	119.7209	118.3646	1.3563 (1.13%)
Flight Time t_f (s)	5.0456	4.9900	4.9771	0.0129 (0.26%)
Peak Height y_{max} (m)	31.2181	30.5075	30.4063	0.1013 (0.33%)
Rod Rotation Analysis				
Net Angle Change $\Delta\theta$ (deg)	—	90.8550	—	—
Total Arc $ \int \omega dt $ (rad)	—	7.8049	—	—
Total Rotations (rev)	—	1.2422	—	—

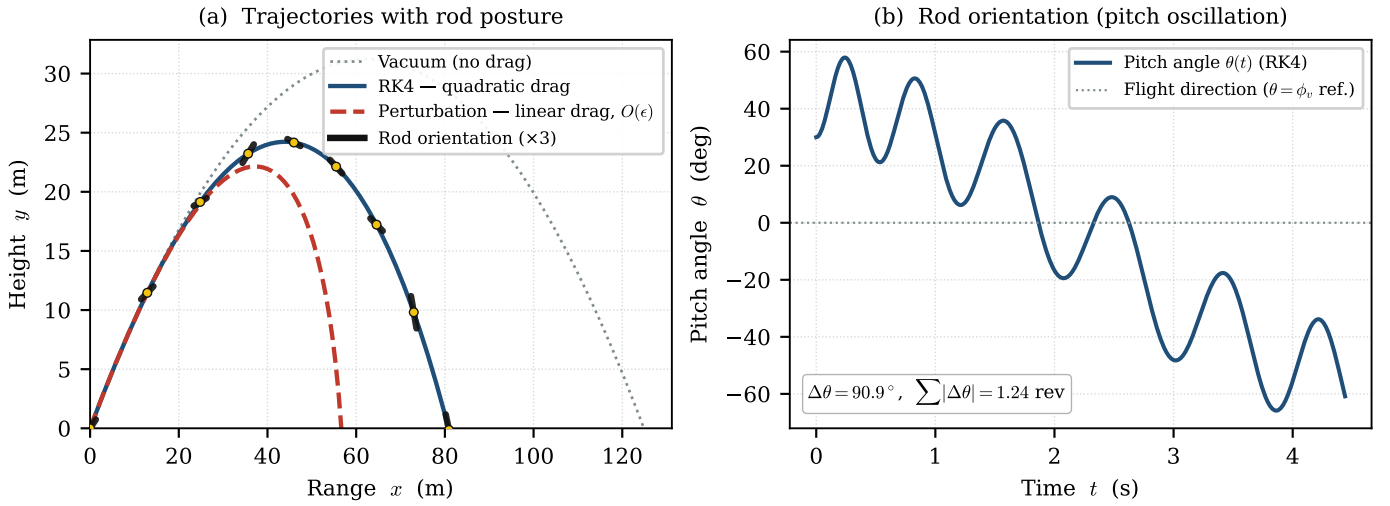


Fig. 1. Rod Mode ($\epsilon = 0.006125 \text{ m}^{-1}$, $\epsilon v_0 T_{vac} = 1.0817$). (a) Centre-of-mass trajectories for the RK4 quadratic-drag model (solid) and the first-order linear-drag perturbation model (dashed), with the frictionless vacuum parabola for reference; black bars show the instantaneous rod orientation sampled along the RK4 path (drawn at $3\times$ the true length for legibility). (b) Evolution of the pitch angle $\theta(t)$, showing a restoring pitch oscillation about the flight-path direction rather than end-over-end tumbling.

a sustained *pitch oscillation* about the instantaneous flight-path direction, with the mean orientation drifting downward as the velocity vector rotates over the descending arc. Over its 4.4400 s flight, the rod's orientation (moment of inertia $I = 0.1667 \text{ kg}\cdot\text{m}^2$) exhibits a net change $\Delta\theta$ of 90.8550° while traversing a total integrated absolute angular path of 7.8049 rad; expressed in revolutions, this accumulated path is equivalent to 1.2422 turns, although the pitch angle itself never exceeds $\pm 90^\circ$ and no full revolution is completed. This continual oscillatory exchange between rotational and translational kinetic energy is a coupled phenomenon completely neglected by the linearized point-mass model.

B. Ball Mode Analysis and Perturbation Validity

In Ball Mode, the physical parameters of the shot put ball ($m = 7.26 \text{ kg}$, $r = 0.06 \text{ m}$, $C_d = 0.47$) yield an area $A = 0.0113 \text{ m}^2$ and an exceptionally low perturbation parameter of $\epsilon = 0.000448 \text{ m}^{-1}$. The unperturbed smallness parameter is

evaluated as $\epsilon v_0 T_{vac} = 0.0792$, which is well below the 0.3 safety limit. Consequently, the weak-drag assumption holds, and the validity warning is bypassed.

As shown in Table I and Fig. 2, the trajectory curves for both models in Ball Mode are virtually identical. The RK4 quadratic drag model predicts a maximum range of 119.7209 m, a flight time of 4.9900 s, and a peak height of 30.5075 m. The first-order perturbation model matches these numbers with remarkable precision, yielding a range of 118.3646 m, a flight time of 4.9771 s, and a peak height of 30.4063 m.

The discrepancy for the ball is only 1.3563 m (1.13%) in range, 0.0129 s (0.26%) in flight time, and 0.1013 m (0.33%) in peak height. The exceptionally high mass-to-area ratio (high ballistic coefficient) of the heavy shot put minimizes the magnitude of the air resistance. Under these conditions, the non-linear coupling terms remain negligible, validating the use of the linearized perturbation expansion as an extremely

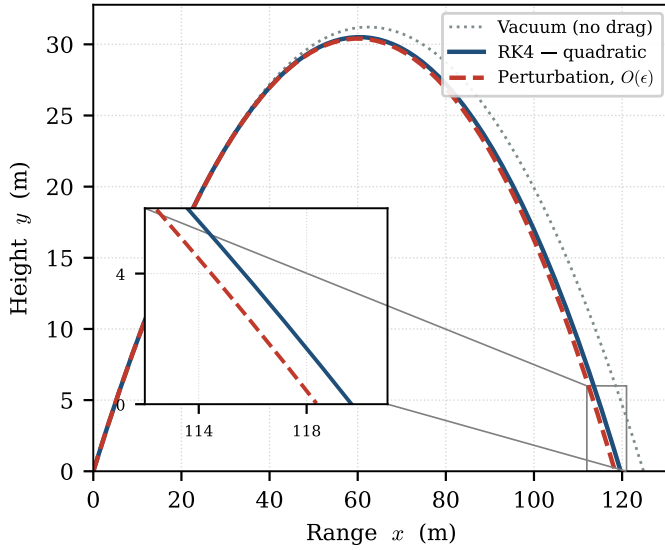


Fig. 2. Ball Mode ($\epsilon = 0.000448 \text{ m}^{-1}$, $\epsilon v_0 T_{vac} = 0.0792$). The RK4 quadratic-drag (solid) and perturbation linear-drag (dashed) trajectories are visually indistinguishable; the inset magnifies the landing region, where the residual 1.13% range difference becomes apparent.

high-accuracy, closed-form analytical surrogate for point-mass projectile tracking.

VI. CONCLUSION

This study highlights the physical and mathematical nuances of projectile motion through a unified, within-system comparative approach. By linearizing the quadratic drag force at the launch speed, we derived a consistent analytical baseline to contrast with high-fidelity RK4 numerical integration. In Ball Mode, where the shot put's high ballistic coefficient maintains a smallness parameter of 0.0792, the linearized first-order perturbation model matches the exact non-linear numerical solution with remarkable fidelity. In Rod Mode, the higher drag-to-mass ratio triggers the simulator's validity warning ($\epsilon v_0 T_{vac} = 1.0817$), revealing that the linearized model overdampens the lower-velocity phases compared to the coupled quadratic integration. Additionally, the rod's non-zero angle of attack at launch generates a $\sin(2\phi)$ aerodynamic pitching moment, leading to a sustained pitch oscillation (a net orientation change of 90.86° and an accumulated angular path equivalent to 1.2422 turns) that can only be captured by the multi-degree-of-freedom RK4 numerical solver. Ultimately, matching the mathematical model to the physical scale of the projectile is vital for ballistic accuracy: a high ballistic coefficient renders the analytical surrogate reliable, while strong drag-to-mass coupling and rotational dynamics demand a full numerical treatment.

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